

# Manual

## **HYDRA Interfacing Module to SAP R/3 PS (CC4) SAP-PSCC4 3.0**

Version 1.1.19800

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# 1 HYDRA Interfacing Module to SAP PS (CC4)

## Overview

### Purpose

You use the SAP-PSCC4 interface to download project orders from SAP PS (networks and their operations) for a specified operation. In the MES, you can record postings for these operations and upload the resulting times to SAP PS. You use the CA-PDC / CC4 interface to connect to SAP PS.

### Implementation notes

You use the function package Interfacing Module to SAP R/3 PS (CC4), if you process projects in SAP PS using networks and if you want to record data for these networks in the MES and then upload this data.

### Integration

If you use the component SAP-PSCC4, the orders/operations transferred with this component are used for a great number of further postings in HYDRA.

### Features

- Transfer of order data
  - Download of the released operations from PS and integration as HYDRA project orders (networks)/project operations (initial and delta download and delete function)
- Recording of times
  - Recording of start/interruption/end of processing and posting of the times according to the HYDRA configuration and validation checks
- Upload of operation postings
  - Upload to SAP PS of the actual times (performances) recorded for a specific SAP time ticket

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## 2 Behavior Depending on the Transfer Types

### Initial download behavior

An initial download of the order and operation data will only take place when the complete system is commissioned and a first database is to be created in HYDRA. In addition, also other scenarios are possible by which the entire operations base will be replaced.

During an initial download all operations from SAP will be transferred that have at least the status "released" or that are not technically completed yet. This means that also (end) confirmed operations will be transferred but not imported in HYDRA.

When the initial download is received, the current operations data base in HYDRA will be deleted and be replaced by the new operations data. The current data will immediately be available.

An initial download will not only delete the order base existing in HYDRA but also all current times and quantities entered for orders and/or transactions.



In addition, all operations that are being executed will not be deleted. This means that they must be interrupted and/or terminated manually in HYDRA.

Conclusion: Any initial download during the operation of HYDRA must be used with greatest care and it would be useful to contact the MPDV Support beforehand.

For security reasons, the initial download function needs to be enabled explicitly as of the below-mentioned program version. Activation is performed by an INI configuration.

.lib\b\_anr.dll

V8.1.1.326

### Delta download behavior

A delta download creates new operations in HYDRA that will then be added to the database. Another function of the delta download is the modification of already transferred operations. Modifications of operations with the status "Running", "Finished" or "Deleted" are not allowed by the HYDRA default settings.



CUSTOMIZING information: Modifications of the operations depend on whether the flag "alterable order data" is set for the corresponding status in the status configuration.

If an operation cannot be modified to ensure the consistency in HYDRA (in general for running operations), this will be logged in HYDRA and be saved for error tracking purposes.

The basis for the delta download in SAP is the database table ORDCOM.

## **Deletion download behavior**

A deletion download deletes those operations that are no longer necessary in the production process from the HYDRA database.

As with the modifications, the deletion result depends on the operation's status. Operations identified as "Running", "Interrupted", "Finished" or "Deleted", will not be deleted.

Moreover, the confirmation number (CONF\_NO) will be checked to identify the orders via order, sequence and transaction. This prevents any accidental deletion of orders that are seemingly the same but which have different confirmation numbers.

## **Behavior when re-importing master data in SAP**

When master data for an existing and released order that has already been transferred to HYDRA are re-imported in SAP this impacts also the interface to HYDRA.

When master data are re-imported, SAP will assign new confirmation numbers for the individual operations even though the order and operation numbers won't change. These are then transferred together with the next delta download to HYDRA. The confirmation/upload number will then be updated in HYDRA.



The update will also be made when the current status is "Running". In this case, however, ONLY the confirmation/upload no. will be changed and all other data (order quantity, scheduling, etc.) will not be updated.

## **Behavior during the technical completion in SAP**

If an order is technically completed in SAP this will lead to a deletion download at the interface. This means that the order data in HYDRA will be deleted.

### 3 Operations Data Download SAP --> HYDRA

#### Operation data

The following structure is part of the PP-PDC / CC4 interface and is used to transfer maintenance orders to subsystems.

Message type: OPERA4

IDoc type: OPERA4

Segments: OPERA4

Message function:

|     |   |                    |
|-----|---|--------------------|
| APP | → | Delta download     |
| DEL | → | Deletion download  |
| UPD | → | Base data download |

| Field Name | T    | L    | Description                   | Rel. KK4 | HYDRA Reference                                    |
|------------|------|------|-------------------------------|----------|----------------------------------------------------|
| RUECK*     | NUMC | 10   | Operation confirmation number | x        | Is stored for the confirmation, not used elsewhere |
| AUFNR      | CHAR | 12   | Order number                  | x        | According to Configuration (*1)                    |
| APLFL      | CHAR | 6    | Operation sequence            | x        | According to Configuration (*1)                    |
| VORNr      | CHAR | 4    | Operation number              | x        | According to Configuration (*1)                    |
| UVORN      | CHAR | 4    | Sub procedure                 | x        | According to Configuration (*1)                    |
| SPLIT*     | NUMC | 3    | Split number                  | x        | Not used                                           |
| KAPAR*     | CHAR | 3    | Capacity type                 | x        | Not used                                           |
| BDEGR*     | CHAR | 3    | Subsystem connection grouping | x        | Not used                                           |
| MGVRG      | DEC  | 10.3 | Default quantity              | -        |                                                    |
| ASVRG      | DEC  | 10.3 | Scrap quantity                | -        |                                                    |
| MEINH      | CHAR | 3    | Operation quantity unit       | -        |                                                    |
| UMREN      | DEC  | 5.0  | Conversion denominator        | -        |                                                    |
| UMREZ      | DEC  | 5.0  | Conversion Numerator          | -        |                                                    |
| KMEIN      | CHAR | 3    | Header quantity unit          | -        |                                                    |
| UNTMG      | DEC  | 10.3 | Underdelivery quantity        | -        |                                                    |
| UEBMG      | DEC  | 10.3 | Overdelivery quantity         | -        |                                                    |
| ACTI1      | DEC  | 10.3 | Planned activity 1            | -        |                                                    |
| UNIT1      | CHAR | 3    | Planned activity 1 unit       | -        |                                                    |
| ACTI2      | DEC  | 10.3 | Planned activity 2            | -        |                                                    |
| UNIT2      | CHAR | 3    | Planned activity 2 unit       | -        |                                                    |
| ACTI3      | DEC  | 10.3 | Planned activity 3            | -        |                                                    |

| Field Name | T    | L    | Description                         | Rel. KK4 | HYDRA Reference                                      |
|------------|------|------|-------------------------------------|----------|------------------------------------------------------|
| UNIT3      | CHAR | 3    | Planned activity 3 unit             | -        |                                                      |
| ACTI4      | DEC  | 10.3 | Planned activity 4                  | -        |                                                      |
| UNIT4      | CHAR | 3    | Planned activity 4 unit             | -        |                                                      |
| ACTI5      | DEC  | 10.3 | Planned activity 5                  | -        |                                                      |
| UNIT5      | CHAR | 3    | Planned activity 5 unit             | -        |                                                      |
| ACTI6      | DEC  | 10.3 | Planned activity 6                  | -        |                                                      |
| UNIT6      | CHAR | 3    | Planned activity 6 unit             | -        |                                                      |
| LMNGA      | DEC  | 10.3 | Planned yield                       | -        |                                                      |
| XMNGA      | DEC  | 10.3 | Planned scrap quantities            | -        |                                                      |
| ISTAT      | CHAR | 5    | Operation progress                  | x        | OPs with completion status ("E") are not transferred |
| ISM01      | DEC  | 10.3 | Actual activity 1                   | -        |                                                      |
| ISM02      | DEC  | 10.3 | Actual activity 2                   | -        |                                                      |
| ISM03      | DEC  | 10.3 | Actual activity 3                   | -        |                                                      |
| ISM04      | DEC  | 10.3 | Actual activity 4                   | -        |                                                      |
| ISM05      | DEC  | 10.3 | Actual activity 5                   | -        |                                                      |
| ISM06      | DEC  | 10.3 | Actual activity 6                   | -        |                                                      |
| LEK01      | CHAR | 1    | Completion indicator activity 1     | -        |                                                      |
| LEK02      | CHAR | 1    | Completion indicator activity 2     | -        |                                                      |
| LEK03      | CHAR | 1    | Completion indicator activity 3     | -        |                                                      |
| LEK04      | CHAR | 1    | Completion indicator activity 4.    | -        |                                                      |
| LEK05      | CHAR | 1    | Completion indicator activity 5     | -        |                                                      |
| LEK06      | CHAR | 1    | Completion indicator activity 6     | -        |                                                      |
| ARBPL      | CHAR | 8    | Workplace                           | x        | OP machine (see note)                                |
| WERKS      | CHAR | 4    | Workplace plant                     | x        | Plant                                                |
| ARBPI      | CHAR | 8    | Actual workplace                    | x        | Not used                                             |
| WERKI      | CHAR | 4    | Actual workplace plant              | x        | Not used                                             |
| ISMNW      | DEC  | 10.3 | Actual work (cannot be negative)    | x        | Not used                                             |
| ISMNE      | CHAR | 3    | Actual work unit                    | x        | Not used                                             |
| ARBEI      | DEC  | 6.1  | Planned work                        | x        | Target duration                                      |
| ARBEH      | CHAR | 3    | Planned work unit                   | x        | "STD" / "H" / "HUR"<br>"MIN"<br>"S" / "SEC"          |
| OFMNW      | DEC  | 6.1  | Remaining work (cannot be negative) | x        | Not used                                             |
| OFMNE      | CHAR | 3    | Remaining work unit                 | x        | Not used                                             |
| LEKNW      | CHAR | 1    | Indicator: no remaining work        | x        | Not used                                             |
| FSAVD      | DATS | 8    | Earliest start date                 | x        | PPS start date                                       |
| FSAVZ      | TIMS | 6    | Earliest start time                 | x        | PPS Start time                                       |
| SSEDD      | DATS | 8    | Latest end date                     | x        | PPS end date                                         |
| SSEDZ      | TIMS | 6    | Latest end time                     | x        | PPS end time                                         |



\*) The actual work (field ISMNW) may not exceed 590 hours.

(\*1) See below for more information on generating the HYDRA order number.

## Upload Requirement

The upload requirement controls the confirmation upload to SAP. All available confirmations will be transferred to SAP in response to such upload requirement sent from SAP to HYDRA.

|                   |                          |
|-------------------|--------------------------|
| Message type:     | REQUI4                   |
| IDoc type:        | REQUI4                   |
| Segments:         | REQUI4                   |
| Message function: | REQ → Upload Requirement |

## 4 Confirmations Upload HYDRA --> SAP

### HYDRA Supported Record Types and Activities

HYDRA BDE reports the following time ticket related record types back to SAP R/3 PP:

| Record Type | SAP Meaning                    | HYDRA triggering action                                                    |
|-------------|--------------------------------|----------------------------------------------------------------------------|
| T20         | Partial Time Ticket Completion | Automatic or manual order interruption at the BDE terminal or BDE console. |
| T40         | Time Ticket Completion         | Order completion message at the BDE terminal or BDE console.               |

#### Please note

When HYDRA MPL is active, each generated output batch (output batch change message) will generate OP interruption or logoff time tickets plus an L20 partial completion message.

### Report Structure E2CONF7

HYDRA ADE reports time ticket related information back to SAP. Its base data consists of the order-related postings (U / E records) and the duration data in these records. The badge number is added before transmission to SAP.

Message type: CONF42  
IDoc type: CONF42  
Segments: E2CONF7

| Field Name | Type | Length | Text                                 | HYDRA Reference |  |  |
|------------|------|--------|--------------------------------------|-----------------|--|--|
| SATZA      | CHAR | 3      | Report record type                   | T20 or T40      |  |  |
| TERID      | CHAR | 4      | Terminal ID                          | Not used        |  |  |
| LDATE      | DATS | 8      | Logical date / Actual reporting date | Logoff date     |  |  |
| LTIME      | TIMS | 6      | Logical time / Actual reporting time | Logoff time     |  |  |
| ERDAT      | DATS | 8      | Report entry date                    | Reporting date  |  |  |
| ERTIM      | TIMS | 6      | Report entry time                    | Reporting time  |  |  |

| Field Name | Type | Length | Text                         | HYDRA Reference                                              |  |  |
|------------|------|--------|------------------------------|--------------------------------------------------------------|--|--|
| BUDAT      | DATS | 8      | Report booking time          | Set based on the booking record shift date                   |  |  |
| ARBPL      | CHAR | 8      | Workplace                    | HYDRA workplace                                              |  |  |
| WERKS      | CHAR | 4      | Plant                        | Plant                                                        |  |  |
| ZAUSW      | NUMC | 8      | Badge number                 | Time badge number                                            |  |  |
| AUFNR      | CHAR | 12     | Order number                 | Follows default                                              |  |  |
| VORN       | CHAR | 4      | Operation number             | Follows default                                              |  |  |
| UVORN      | CHAR | 4      | Sub-operation number         | Follows default                                              |  |  |
| SPLIT      | NUMC | 3      | Split number                 | Not used                                                     |  |  |
| KAPAR      | CHAR | 3      | Capacity type                | Not used                                                     |  |  |
| ABWEI      | CHAR | 4      | Deviation basis              | Not used                                                     |  |  |
| ABARB      | NUMC | 3      | Completion level             |                                                              |  |  |
| PEDD       | DATS | 8      | Predicted end date           | Not used                                                     |  |  |
| PEDZ       | TIMS | 6      | Predicted end time           | Not used                                                     |  |  |
| LEKNW      | CHAR | 1      | Indicator: no remaining work | Not used                                                     |  |  |
| LTXA1      | CHAR | 40     | Report text                  | Not used                                                     |  |  |
| ISMNW *    | DEC  | 6.1    | Actual work                  | P_DAUER                                                      |  |  |
| ISMNE      | CHAR | 3      | Actual work unit             | "H" / "STD" / "HUR"<br>"MIN"<br>"S" / "SEC"<br>Default "STD" |  |  |
| LEARR      | CHAR | 6      | Activity type                | Not used                                                     |  |  |
| IDAUR *    | DEC  | 4.1    | Actual duration              | Actual duration BMK11                                        |  |  |
| IDAUE      | CHAR | 3      | Actual duration unit         | "H" / "STD" / "HUR"<br>"MIN"<br>"S" / "SEC"<br>Default "STD" |  |  |
| ODAU       | DEC  | 4.1    | Remaining duration           | Not used                                                     |  |  |
| ODAUE      | CHAR | 3      | Remaining duration unit      | Not used                                                     |  |  |
| OFMNW *    | DEC  | 6.1    | Remaining work               | Not used                                                     |  |  |
| OFMNE      | CHAR | 3      | Remaining work unit          | Not used                                                     |  |  |
| ISDD       | DATS | 8      | Execution start date         | Logon date                                                   |  |  |
| ISDZ       | TIMS | 6      | Execution start time         | Logon time                                                   |  |  |
| IEDD       | DATS | 8      | Execution end date           | Logoff date                                                  |  |  |
| IEDZ       | TIMS | 6      | Execution end time           | Logoff time                                                  |  |  |

## 5 MYERPRCK - Program Parameters

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

### Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

### Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

| Parameters                                | Meaning/use                                                                                                                                                                                                    | Relevant interfaces                                      | Productive release      |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------|
| Program parameters to control processing: |                                                                                                                                                                                                                |                                                          |                         |
| /MESTYP=XXXX                              | The parameter MESTYP defines the structure to be generated.                                                                                                                                                    | All                                                      | Yes                     |
| /GRP=XXXX                                 | The grouping type specifies the criterion by which uploads should be grouped.<br>Possible values:<br><br>PLANT --> Groups by plant                                                                             | Requires customizations                                  | Requires customizations |
| /V=sssss                                  | Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time.<br><br>Use the parameter /V=sssss (sssss = | EIS-ERP<br><br>EIS-XPPS<br><br>SAP-PPDC<br><br>SAP-PPREM | Yes                     |

| Parameters                                                                                                                               | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Relevant interfaces                                                                                                                | Productive release |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                                                                                                                                          | <p>delay time in seconds) to activate the above described delay when the upload program is called.</p> <p>Examples:</p> <pre>myerprck.exe/out /V=3600</pre> <p>The system only uploads postings that are older than one hour.</p>                                                                                                                                                                                                                                                                                                                                         | <p>SAP-PPPI</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> <p>SAP-COILV</p>                                                                 |                    |
| <p><b>/BIS=DDMMYYHHMM</b></p> <p><b>/BIS=HHMM</b></p> <p><b>/TILLDATE=MM/DD/YYYY</b></p> <p><b>/TILLTIME=sec</b>      after midnight</p> | <p>Use the parameter <b>/BIS=DDMMYYHHMM</b> (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day.</p> <pre>Myerprck.exe /BIS=2505110600</pre> <p>This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011.</p> <pre>Myerprck.exe /BIS=0600</pre> <p>This parameter uploads postings that were recorded until 06:00 a.m. of the current day.</p> | <p>EIS-ERP</p> <p>EIS-XPPS</p> <p>SAP-PPDC</p> <p>SAP-PPREM</p> <p>SAP-PPPI</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> <p>SAP-COILV</p> | Yes                |
| <b>/TZ=+/-sssss</b>                                                                                                                      | <p>Use the parameter <b>/TZ=+/-sssss</b> to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPDC interface according to its specifications.</p>                                                                                                                                                                                                                                                                                 | SAP-PPDC                                                                                                                           | Yes                |

| Parameters | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Relevant interfaces                                                                             | Productive release |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
| /KST=XXX   | <p>Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center.</p> <p>Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine.</p> <p>You can specify the parameter several times per call.</p> <p><u>Example:</u></p> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> <p>The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /CLEAR_RES | Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SAP-PPDC                                                                                        | Yes                |
| /NEG_MENGE | By default, quantities (L20/L40) cannot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SAP-PPDC                                                                                        | Yes                |

| Parameters      | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Relevant interfaces   | Productive release |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|
|                 | <p>be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPDC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished.</p> <p>This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.</p> |                       |                    |
| /LA_MNR         | <p>The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM.</p> <p>The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type.</p> <p>Then the system uses the machine to identify the activity type from the <a href="#">activity types</a> kept in HYDRA.</p>                                                                                                                                                                                                          | SAP-PMCC3             | Yes                |
| /IDENT_PRAEFIX= | <p>In the upload structure of the SAP-PPDC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field.</p> <p>You can add a prefix to the EX_IDENT field to differentiate between uploads</p>                                                                                                                                                                                                                                                                                                                                                                   | SAP-PPDC<br>SAP-PPDCC | Yes                |

| Parameters  | Meaning/use                                                                                                                                                                                                                               | Relevant interfaces                                                                             | Productive release |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
|             | <p>from various HYDRA subsystems connected to one SAP instance.</p> <p><u>Example:</u></p> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> <p>The prefix may only include hexadecimal characters: A –H und 0 – 9.</p>                          |                                                                                                 |                    |
| /ABZEICH=XX | <p>While customizing the order type, you can specify that only signed data records are uploaded.</p> <p>Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.</p>       | EIS-ERP<br>EIS-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /TRANSFER=  | <p>Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system.</p> <p>The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |



| Parameters       | Meaning/use                                                                                                                                                                                                                                                                      | Relevant interfaces                                                                              | Productive release |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------|
| /NOTTRANSFER=XXX | <p>Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system.</p> <p>The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.</p>                                 | EIS-ERP<br>ESI-XPPS<br>SAP-PPPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /SEK             | <p>The EIS-ERP interface uploads the times of resource performance accounts in hours.</p> <p>In particular with very short lead times this may effect that logon times are cut off by a conversion into hours.</p> <p>Use this program parameter to upload times in seconds.</p> | EIS-ERP<br>ESI-XPPS                                                                              | Yes                |
| /RMTYP=          | <p>When customizing the order type, you can assign an upload type to the order type.</p> <p>Use this program parameter to only upload data records of this upload type.</p> <p>You can specify the parameter several times per call.</p>                                         | EIS-ERP<br>ESI-XPPS<br>SAP-PPPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |

| Parameters | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Relevant interfaces                                                                             | Productive release      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| /KAT=      | <p>When customizing the order type, you can connect the order type with a category.</p> <p>Use the program parameter /KAT= to only upload data records of this category.</p> <p>You can specify the parameter several times per call.</p>                                                                                                                                                                                                                               | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                     |
| /SART=     | <p>The system only uploads ADE log postings of the specified record type.</p> <p>Therefore, you can use different program parameters per call and record type for uploading.</p> <p>Requirement: You have to activate the corresponding uploads when customizing the order type.</p> <p>You can specify the parameter several times per call.</p> <p><u>Example:</u></p> <pre>Myerprck.exe      /SART=A /SART=E</pre> <p>⇒ The system only uploads A and E records.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                     |
| /NOLOCK    | <p>When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,</p>                                                                                                                                                                                                                                                                                                                     | All                                                                                             | Requires customizations |

| Parameters    | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                        | Relevant interfaces               | Productive release |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------|
|               | <p>the upload is not carried out.</p> <p>You can use this program parameter to prevent this check.</p> <p>Set this parameter, in particular, if the upload is not based on the table ade_protokoll.</p>                                                                                                                                                                                                                            |                                   |                    |
| /EINH_CC34    | <p>The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter <b>"/EINH_CC34"</b> to upload the data in other units. The following units are supported:</p> <p>Hours: H, HUR, STD</p> <p>Minutes: MIN</p> <p>Seconds: SEC</p> <p><u>Example:</u></p> <pre>Myerprck.exe /EINH_CC34=HUR</pre> <p>The system uploads the recorded times in the unit "HUR" (hours).</p> | <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> | Yes                |
| /SDAT_STORNO  | <p>The SAP-PPPDCC interface transfers the change date along with the correction records.</p> <p>Use this program parameter to upload the initially collected shift date instead.</p>                                                                                                                                                                                                                                               | SAP-PPPDCC                        | Yes                |
| /NORFC_STORNO | <p>The SAP-PPPDCC interface transfers the cancellation records via <a href="#">sRFC</a>.</p> <p>Use the program parameter to transfer the data in the IDoc format to SAP. To do</p>                                                                                                                                                                                                                                                | SAP-PPPDCC                        | Yes                |

| Parameters       | Meaning/use                                                                                                                                                                                                                                                                                                     | Relevant interfaces                               | Productive release      |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|
|                  | <p>so, inbound processing must be implemented in SAP.</p> <p>The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.</p>                                                                                            |                                                   |                         |
| /PI              | <p>If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly.</p> <p>Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).</p> | <p>SAP-PPDC</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> | Yes                     |
| /INDEX_TMP_TABLE | <p>Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected.</p> <p>To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.</p>                                                                  | All                                               | Requires customizations |
| /UE_PARAMS=      | Program parameter for the stand-alone user exit processing (DD format).                                                                                                                                                                                                                                         | Various                                           | Yes                     |
| /NOSTORNO        | <p>Use this program parameter to prevent cancellation records from being uploaded.</p> <p>Therefore, you can use different program parameters per call and record type for uploading.</p> <p>Requirement: You have to activate the</p>                                                                          | All                                               | Yes                     |

| Parameters                                          | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                | Relevant interfaces | Productive release      |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------|
|                                                     | corresponding uploads when customizing the order type.                                                                                                                                                                                                                                                                                                                                                                                     |                     |                         |
| /RECALC_NEG_YIELD                                   | Use this parameter to offset negative yield with already posted positive uploads.                                                                                                                                                                                                                                                                                                                                                          | SAP-PPPDCC          | Requires customizations |
| Program parameters to use the SIGUSR communication: |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                         |
| /LOGGING                                            | Use this program parameter to activate communication from the database table HYD_LOGGING.<br><br>To do so, a customization might be required.                                                                                                                                                                                                                                                                                              | INDIVIDUAL CASE     | Yes                     |
| /WAIT_SIGUSR1=XX                                    | The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.                                                                                                                                                                                                                                                                                     | INDIVIDUAL CASE     | Yes                     |
| /PEEK_SIGUSR1=XX                                    | Use this parameter to delay execution of an action triggered by the SIGUSR communication.<br><br>The delay time is entered in seconds for this parameter.<br><br>The program interprets this time as follows:<br><br>If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds.<br><br>If in a specific case, triggers would indeed arrive every second then the | INDIVIDUAL CASE     | Yes                     |

| Parameters        | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Relevant interfaces | Productive release |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|
|                   | WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                    |
| /SEND_SIGUSR1=    | <p>This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication.</p> <p>Specify the process/program WITHOUT file extension.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | INDIVIDUAL CASE     | Yes                |
| /COUNT_SIGUSR1=XX | <p>Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance.</p> <p>Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1.</p> <p>Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to</p> | INDIVIDUAL CASE     | Yes                |

| Parameters                                                            | Meaning/use                                                                                                                                                                                                                                                   | Relevant interfaces | Productive release |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|
|                                                                       | <p>the DB.</p> <p>Benefits:</p> <p>If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.</p> |                     |                    |
| Program parameters for debugging/ tracing/ testing/ logging purposes: |                                                                                                                                                                                                                                                               |                     |                    |
| /ONLYERR                                                              | <p>This program parameter specifies that system log entries are only created if an error occurred during uploading.</p> <p>This reduces the entries in the system log.</p>                                                                                    | All                 | Yes                |
| /SIM                                                                  | The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").                                                                                                                                               | All                 | No                 |
| /SIMULATION                                                           | The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).                                                                                                                                          | All                 | No                 |

## 6 Application-Relevant Settings in HYDRA

### Maintenance of the HYDRA distribution model – inbound processing

Edit entries for HYDRA inbound processing in the [HYDRA distribution model](#):

| Parameter name                   | Value                                                |
|----------------------------------|------------------------------------------------------|
| For processing of network plans  |                                                      |
| Message type                     | OPERA4                                               |
| Priority                         | None                                                 |
| Command                          | mle72imp.scr                                         |
| Command parameter                | /VARIANTE=< <a href="#">MLE variant to be used</a> > |
| Description                      | PS-CC4 – Download of network plans                   |
| Log. target system               | Created logical system                               |
| Storage duration                 | 10                                                   |
| For processing of upload request |                                                      |
| Message type                     | REQUI4                                               |
| Priority                         | High                                                 |
| Command                          | hysapupl.scr                                         |
| Command parameter                | /UPLSEGNAM=E2CONF7                                   |
| Description                      | PS-CC4 – Upload request                              |
| Log. target system               | Created logical system                               |
| Storage duration                 | 10                                                   |

### Maintenance of the HYDRA distribution model – outbound processing

Edit entries for the HYDRA outbound processing in the [HYDRA distribution model](#):



| Parameter name             | Value                           |
|----------------------------|---------------------------------|
| For uploading time tickets |                                 |
| Message type               | CONF42                          |
| Description                | PS-CC4 – Upload of time tickets |
| IDoc type                  | CONF42                          |
| Storage duration           | 10                              |
| Log. target system         | Created logical system          |
| Segment name 1             | E2CONF7                         |

## Activation of initial download

As of program version

.lib\b\_anr.dll V8.1.1.326

the initial download has to be enabled explicitly for security reasons.

Create the following entry in [HYDRA INI configuration](#) if you would like to activate the initial download for the system:

| Parameter name | Value                                 |
|----------------|---------------------------------------|
| INI name       | SAP                                   |
| Section        | INITIAL_DOWNLOAD_ACTIVATION           |
| Key            | ACTIV_TILL                            |
| Value          | <Date value in the format MM/DD/YYYY> |
| Active         | Yes                                   |
| Comment        | Activation of initial download        |

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## 7 Protecting fields of planned operations

### Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

### Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

### Procedure from service pack 12 onwards (b\_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

| Field name  | Value                |
|-------------|----------------------|
| Name        | BAPINOUPDATE         |
| Description | Enter a description. |

For this entry, create an entry including the following values in [INI data configuration](#):

| Field name | Value                                                                                        |
|------------|----------------------------------------------------------------------------------------------|
| Section    | ANR                                                                                          |
| Key        | List the fields (HYDRA BAPI acronyms) that are not overwritten.                              |
| Value      | The value includes a condition. Enter the condition, for example, as follows:<br>ANR.ATYP=AG |
| Active     | Yes                                                                                          |

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

## Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

| Field name  | Value                |
|-------------|----------------------|
| Name        | BAPINOUPDATE         |
| Description | Enter a description. |

For this entry, create an entry including the following values in [INI data configuration](#):

| Field name | Value                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| Section    | ANR                                                                                                                      |
| Key        | List the fields (HYDRA BAPI acronyms) that are not overwritten.                                                          |
| Value      | Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value. |
| Active     | Yes                                                                                                                      |

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

| Database field      | Values/content                                                             |
|---------------------|----------------------------------------------------------------------------|
| INI_VERWEIS         | The value of the VERWEIS column identified from the HYD_INI table via SQL. |
| SECTION             | Section                                                                    |
| IDENT               | Key                                                                        |
| VALUE               | Value                                                                      |
| BEMERKUNG (comment) | Comment                                                                    |
| AKTIV               | Active                                                                     |

## List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

| Acronym      | Meaning                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------|
| ANR.MGRP     | Machine group                                                                                   |
| ANR.MNR      | Workplace/<br>machine                                                                           |
| ANR.OPT:PLAN | Planning indicator:<br>M      Planned for workplace/machine<br>G      Planned for machine group |
| ANR.DATB     | Start date planned (via HLS)                                                                    |
| ANR.ZEIB     | Start time planned (via HLS)                                                                    |
| ANR.DATE     | End date planned (via HLS)                                                                      |
| ANR.ZEIE     | End time planned (via HLS)                                                                      |

## Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

| Field name | Value                          |
|------------|--------------------------------|
| Section    | ANR                            |
| Key        | ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@ |
| Value      | ANR.ATYP=AG@ANR.OPT:PLAN=M@    |
| Active     | Yes                            |

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

## Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

| Field name | Value                                |
|------------|--------------------------------------|
| Section    | ANR                                  |
| Key        | ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@ |
| Value      | ANR.ATYP=AG@ANR.OPT:PLAN=M@          |
| Active     | Yes                                  |

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```